

Here, -s => source IP address of the packet.

The above command will insert a rule to the input chain of the filter table at the second position. This rule will reject all the packets coming from the IP address 10.10.6.203/24.

To append a rule to a table, use this syntax:

```
sudo iptables -A <target_chain> -t <target_table> <YOUR_RULE>
-j <action_name>
```

Here, -A=> add/append. The rule is added at the bottom of target chain.

To give an example:

```
sudo iptables -A INPUT -t filter -d 10.10.6.203/24 -j REJECT
```

Here, -d => destination IP address of the packet.

The above command will add an entry at the end of an input chain of the filter table. This rule will block all the packets going to 10.10.6.203/24

To delete a rule, use this syntax:

```
sudo iptables -D <target_chain> <rule_position> -t <target_table>
```

Here, -D => delete.

As an example:

```
sudo iptables -D INPUT 3 -t filter
```

The above command will remove the third rule from the input chain of the filter table.

To flush/delete all the rules, use the syntax given below:

```
sudo iptables -F -t <target_table>
```

Here, -F => flush/remove all rules.

To give an example:

```
sudo iptables -F -t filter
```

The above command will flush all the rules from the filter table.

To block a protocol, use the following syntax:

```
sudo iptables -A <target_chain> -t <target_table> -p
<protocol_name> -j REJECT
```

Here, -p => protocol name.

As an example:

```
sudo iptables -A INPUT -t filter -s 10.10.6.203 -p tcp -j
REJECT
```

The above command will discard all TCP packets arriving

from 10.10.6.203.

To block all protocols from 10.10.6.203, change the protocol name from 'tcp' to 'all'.

To block a TCP service, the TCP port name/port number associated with that TCP service needs to be mentioned in the command, which will open/close that port for communication.

As an example:

```
sudo iptables -A INPUT -t filter -s 10.10.6.203 -p tcp --dport
22 -j REJECT
```

Here, --dport => destination port.

The above command will close TCP port 22, which offers the SSH (Secure Shell) service. So, all the packets coming from 10.10.6.203, seeking the SSH service from the host, will be rejected.

Saving the rules

Whenever rules are created, they are saved into the volatile memory of the system. If a system is restarted, all the IPTables rules may get flushed unless they are saved. So, it is necessary to save all the rules in order to make them permanent.

The commands used to save IPTables rules vary with different Linux distributions.

For CentOS and Fedora, the command is:

```
/etc/init.d/iptables save
```

For Ubuntu, use:

```
iptables-save> /etc/iptables.rules
```

For all other distributions, use:

```
iptables-save> /etc/sysconfig/iptables
```

With each system restart, these commands will invoke all previously saved IPTables rules.

Summary

Firewalls are a crucial component of any computer network, but they are never adequate. One should not rely only on firewalls that deal with security issues like corrupted packets and network intrusions. Regular data backups, secure password policies, and restricted public access to servers will also safeguard your network. Hopefully, this article will help you to create your own firewall using IPTables that will protect your system against malignant attacks. 

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